

BACS2093 Operating Systems (Practical Assignment) AY 202505

<u>Course Learning</u>	CLO3: Build complex programs in LINUX/UNIX Shell scripting. (P3) (PLO3)
<u>Team Organization</u>	This is a group assignment. Students are to work in a team of 2-3 members. Even though this is a team assignment, each team member should participate in preparing every deliverable required in the assignment and the marks will be given based on each individual contribution.
<u>Academic Impropriety</u>	Your attention is drawn to the University Guidelines on Avoiding Plagiarism. This covers cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment. The work you submit must conform to these regulations. This publication is available on TARUMT Intranet: https://portal.tarc.edu.my/portal/login.jsp
<u>Late Submission</u>	With the exception of EMC reasons, penalty for late submission of assignment shall be imposed after the submission deadline / extended submission deadline: i) Late submission within 1 - 3 days: total marks to be deducted is 10 marks. ii) Late submission within 4 - 7 days: total marks to be deducted is 20 marks. iii) Late submission after 7 days: reject coursework and zero mark shall be awarded. For any late submission of assignment, student shall complete the Late Submission of Coursework Form (as attached) and attach it together with their coursework. The lecturer/tutor shall also acknowledge receipt in the form.
<u>Submission Deadlines</u>	Week 12

**** Important Note: All tasks below should provide comprehensive validations.**

Task 1:

Create a menu script that has the following selections (Figure 1):

Equipment Maintenance Menu

A – Add New Computer Lab Equipment Details
 S – Search Equipment by Serial Number
 U – Update an Equipment Details
 D – Delete an Equipment Details
 M – Sort Equipment by Model
 T – Sort equipment by Status
 P – Sort equipment by Type

Q – Exit from Program

Please select a choice:

Figure 1: Equipment Maintenance Menu

Task 2:

If the user has selected “A” to “**Add New Equipment Details**”, the system should request the lab officer to enter the equipment details. The output of the screen should be similar to Figure 2.

Add Equipment Details Form

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Equipment ID:

Type:

Model:

Serial Number:

Status (Available / Unavailable):

Purchase Date (MM-DD-YYYY):

Warranty Date (MM-DD-YYYY):

Press (q) to return to Equipment Maintenance Menu.

Add another new Equipment details? (y)es or (q)uit :

Figure 2: Add New Equipment Details Form

Note:

1. There are many types of equipment: laptop, monitor, mouse, keyboard, etc.
2. Set the current date as default value for “Purchase Date” input field.

The output of the Equipment details should be stored in the **Equipment.txt** file. Use “:” as field separator. Refer to Sample 1.

ID>Type:Model:Serial:Status:PurchaseDate:WarrantyDate
1001:Laptop:Dell Latitude 7420:SN123456789:Available:2025-01-15: 2026-01-15
1002:Desktop:HP EliteDesk 800 G6:SN987654321:Unavailable:2025-06-10:2027-06-10

Sample 1: Equipment.txt

Task 3:

If the user has selected “S”, the system should redirect the user to “**Search Equipment by Serial Number**”. The output of the screen should be similar to Figure 3.

Search Equipment

Enter Serial Number: SN123456789

Equipment ID: 1001
Type: Laptop
Model: Dell Latitude 7420
Status (Available / Unavailable): Available
Purchase Date (YYYY-MM-DD): 2025-01-15
Warranty Date (YYYY-MM-DD): 2026-01-15

Press (q) to return to Equipment Maintenance Menu.

Search another Equipment? (y)es or (q)uit :

Figure 3: Search Equipment Details

Task 4:

If the user has selected “U”, the system should redirect the user to “**Update Equipment Details**”. The output of the screen should be similar to Figure 4.

Update Equipment Details
Enter Equipment ID: 1001
Serial Number: SN123456789 Type: Laptop Model: Dell Latitude 7420 Status (Available / Unavailable): Unavailable Purchase Date (YYYY-MM-DD): 2025-01-31 Warranty Date (YYYY-MM-DD): 2027-12-31
Press (q) to return to Equipment Maintenance Menu. Are you sure you want to UPDATE the above Equipment Details? (y)es or (q)uit:

Figure 4: Update Equipment Details

Note:

The details can be updated by Equipment such as **Status**, **Purchase Date**, and **Warranty Date**. After the Equipment details has been updated, the system should be able to update it from the **Equipment.txt** file. Refer to Sample 2 below:

ID>Type:Model:Serial:Status:PurchaseDate:WarrantyDate 1001:Laptop:Dell Latitude 7420:SN123456789:Unavailable:2025-01-31:2027-12-31 1002:Desktop:HP EliteDesk 800 G6:SN987654321:Unavailable:2025-06-10:2027-06-10

Sample 2: Equipment.txt

Task 5:

If the user has selected “D”, the system should redirect the user to “**Delete Equipment Details**”. The output of the screen should be similar to Figure 5.

Delete Equipment Details
Enter Equipment ID: 1001
Serial Number: SN123456789 Type: Laptop Model: Dell Latitude 7420 Status (Available / Unavailable): Unavailable Purchase Date (YYYY-MM-DD): 2025-01-31 Warranty Date (YYYY-MM-DD): 2027-12-31
Press (q) to return to Equipment Maintenance Menu. Are you sure you want to DELETE the above Equipment Details? (y)es or (q)uit:

Figure 5: Delete Equipment Details

Note:

After the equipment details has been deleted, the system should be able to remove it from the **Equipment.txt** file. Refer to Sample 3 below:

ID:Type:Model:Serial:Status:PurchaseDate:WarrantyDate
1002:Desktop:HP EliteDesk 800 G6:SN987654321:Unavailable:2025-06-10:2027-06-10

Sample 3: Equipment.txt**Task 6**

For Task 6, each member is required to choose one of the following:

- (i) If the user has selected “M” to **Sort Equipment by Model** (ascending order), the output of the screen should be similar to Figure 6 and the output should be able to generate as a normal text file.

Equipment Details Sorted by Model					
Model	Equipment ID	Type	Serial Number	Status	Warranty Date
HP EliteDesk 800 G6	1001	Desktop	SN987654321	Unavailable	2027-06-10
Press (q) to return to Equipment Maintenance Menu.					
Would you like to export the report as ASCII text file? (y)es (q)uit:					

Figure 6: Sort Equipment by Last Name in Ascending Order

- (ii) If the user has selected “T” to **Sort Equipment by Status**, the output of the screen should be similar to Figure 7 and the user should enter the status either **Available** or **Unavailable**. The output screen should be able to generate as a normal text file.

Equipment Details Sorted by Status					
Enter Equipment Status: Unavailable					
Model	Equipment ID	Type	Serial Number	Purchase Date	Warranty Date
HP EliteDesk 800 G6	1001	Desktop	SN987654321	2025-06-10	2027-06-10
Press (q) to return to Equipment Maintenance Menu.					
Would you like to export the report as ASCII text file? (y)es (q)uit:					

Figure 7: Sort Equipment by Equipment ID in Ascending Order

- (iii) If the user has selected “P” to **Sort Equipment by Equipment Type**, the output of the screen should be similar to Figure 8 and the output should be able to generate as a normal text file.

Equipment Details Sorted by Equipment Type					
Enter Equipment Type: Desktop					
Model	Equipment ID	Status	Serial Number	Purchase Date	Warranty Date
HP EliteDesk 800 G6	1001	Unavailable	SN987654321	2025-06-10	2027-06-10
Press (q) to return to Equipment Maintenance Menu.					
Would you like to export the report as ASCII text file? (y)es (q)uit:					

Figure 8: Sort Equipment by Equipment Type

~END~